

Development of a Defect Estimation Method for CFRP Interstage Structures with Holes using Finite Element Analysis and Graph Neural Networks

Keisuke Nishioka¹ Yuta Kojima² Toshiya Saito³ Masahito Washiya³ Kosuke Kawakami³ Mayu Muramatsu⁴

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¹Keio University, Sci. for Open & Environ. Systems ²Nagoya University

³JAXA, R&D Directorate ⁴Keio University, Mech. Engineering

Motivation: defect localization in CFRP interstage structures

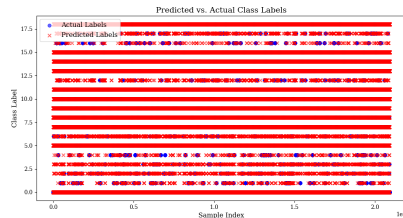
CFRP carries primary load in reusable space transportation (high strength-to-weight).

Interlaminar delamination around bolt holes critically degrades integrity.

Infrared stress measurement (NDT) gives a full-field surface map of the *sum of principal stresses* (DSPSS) [1, 2]; sub-surface defects are inferred from it.

Task

Per-node classification into 19 classes — *in-plane defect region* $\times$ *insertion layer* — directly on the FEA surface mesh.



Predicted vs. true node labels on the perforated specimen.

Prior work and scope of this talk

- **Prior framework** (Frontiers in Materials, 2025) [3]: a GAT-based GNN localizes defects in perforated CFRP from FEA-simulated DSPSS, with *difference-normalized* inputs to suppress hole stress concentration.
- **This talk** extends it along two axes:
 1. robustness under *experimentally-motivated noise* + defect-free negatives;
 2. a controlled **architecture study** — from attention GNNs to *mesh-physics* GNNs and four novel MeshGraphNet variants.

Data generation

Synthetic FEA under tensile loading; interlaminar delamination modeled explicitly by **Teflon inserts** between selected plies [3].

This minisymposium is dominated by *PDE-solver / structure-preserving* methodology. Our contribution is the only **real-structure NDT pipeline**:

- vs. **thermodynamics-preserving GNNs** (Cueto et al.) [4, 5] — they solve/constrain physics; we *apply* GNNs to defect inference on real geometry.
- vs. **momentum-conserving physics-informed GNNs** (Sharma & Fink) — dynamics/conservation; we address static thermoelastic DSPSS.
- vs. **Fourier-embedded PINN $\times$ DIC** (Liu & Yi) [6] — full-field experimental measurement; shared sim-to-real motivation.

Three differentiators

(1) real engineering structure & NDT pipeline; (2) extreme class imbalance (defects < 3%); (3) baseline-free & OOD generalization to unseen defect sizes.

One specimen → one graph

- 13,942 nodes (two stacked plies around the hole)
- Node features: $[x, y, z, \text{DSPSS}]$
- Edges: FEA mesh connectivity, with coordinate-difference edge features $[dx, dy, dz, \|\cdot\|]$
- 19 classes (region $\times$ layer); class 0 = defect-free

Input = *difference-normalized* DSPSS (damaged – pristine) [3], isolating the defect signal from hole stress concentration.

The imbalance trap

Class-0 prior ≈ 0.998 . Predicting “defect-free” everywhere gives 99.8% accuracy yet **macro-F1 ≈ 0** .

⇒ report **macro-F1**; engineer the training distribution, not only the loss.

Handling extreme class imbalance

The recipe that makes learning possible:

1. **Noise-augmented difference data** — robustness to acquisition noise [7].
2. **Defect-free negatives (NDF)** — 5000 genuine class-0 graphs (train only).
3. **Balanced sampling 5000:5000 + Focal loss** [8] (cf. LDAM [9], logit adjustment [10], GraphSMOTE [11]).
4. Leakage control: group-purged val/test split by specimen.

Dominant factor

With the recipe, a faithful reproduction reaches macro-F1 $\approx$ **0.70**, vs. ≤ 0.08 without it — the *data distribution* dominates.

Model zoo: from attention to mesh-physics GNNs

Model	Mechanism	
GAT [12]	attention over neighbours (Frontiers baseline).	
GATv2 [13]	<i>dynamic</i> attention (strictly more expressive).	
GraphSAGE	max-aggregation of neighbour features.	All use
PNA	multiple aggregators + degree scalers.	
MeshGraphNet [14]	Encode-Process-Decode; node <i>and</i> edge updates each step [15].	

coordinate-based geometric features (NDT-admissible). Equivariant GNNs [16] considered as future structure-aware option.

Architecture sweep under the balanced recipe

Identical recipe, 120 epochs, balanced 5000:5000, group-purged eval.

Architecture	macro-F1	note
MeshGraphNet [14]	0.764	Encode-Process-Decode, edge update
GATv2 [13]	0.694	dynamic attention
GAT [12]	0.674	Frontiers baseline (~0.61 published)
GraphSAGE	0.663	max-aggregation
PNA	0.613	multi-aggregator

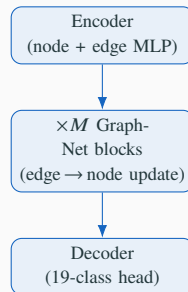
Takeaway

Under the balanced recipe, the **mesh-physics architecture** wins — beating the legacy GAT recipe (0.70) and the published baseline (0.61).

Why MeshGraphNet fits the problem

Encode–Process–Decode [14, 15]:

- node *and* edge embeddings updated at every message-passing step (residual);
- edge feature = coordinate difference
 $[dx, dy, dz, \text{dist}] \Rightarrow$ local geometry & anisotropy;
- matches FEA-mesh stress-propagation structure better than attention-only GNNs.



Novelty: four MeshGraphNet variants for layered NDT

Variant	Idea / why it could matter for CFRP
MeshGraphKAN	Fourier-KAN node encoder — learnable spectral basis for the <i>high-frequency stress-concentration</i> field (spectral-bias mitigation) [6].
Hybrid-MGN	mesh edges + <i>front/back cross-layer “world” edges</i> — structurally separates the two plies.
BiStride-MGN	per-graph global-context broadcast — multiscale, long-range stress propagation.
MGN-Transformer	interleaved per-graph self-attention — global node interaction.

Hypothesis

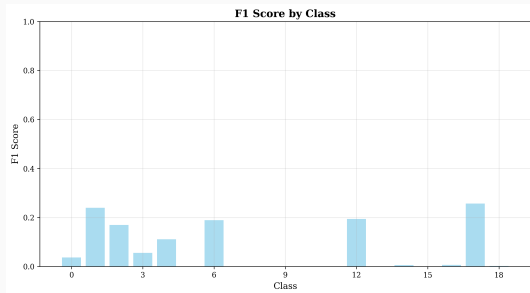
MeshGraphKAN × **stress concentration** and **Hybrid-MGN** × **two-ply separation** are the two novel angles most likely to cross macro-F1 0.80.

Variant	macro-F1	vs. MeshGraphNet (0.764)
MeshGraphNet (baseline)	0.764	—
MeshGraphKAN	TBD	
Hybrid-MGN	TBD	
BiStride-MGN	TBD	
MGN-Transformer	TBD	

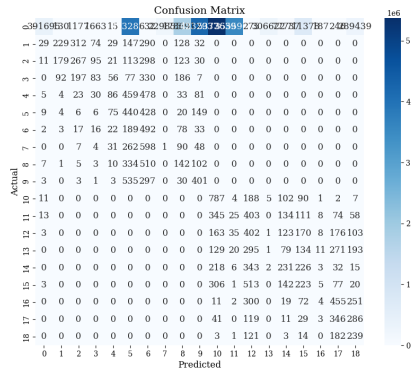
Target

Cross the **0.80** macro-F1 line with a physically-motivated variant.

Per-class performance & confusion



Per-class F1.



Confusion matrix (top classes).

- The **data recipe** (noise-aug difference + NDF negatives + balancing) is dominant: 0.08 $\rightarrow$ 0.70.
- Under the balanced recipe, **MeshGraphNet** [14] is the best architecture (**0.764**), beating the published GAT baseline (0.61).
- Two **novel layered-NDT variants** (Fourier-KAN, front/back Hybrid) target > 0.80 .

Future work

- **Sim-to-real:** transfer to experimental infrared-stress (TSA) data.
- **Physics consistency:** structure-preserving GNNs [4].
- **Application:** aerospace payload-fairing structural health monitoring.

- [1] J. M. Dulieu-Barton and P. Stanley. **“Development and applications of thermoelastic stress analysis”**. In: *The Journal of Strain Analysis for Engineering Design*, 33(2), 93-104 (1998). DOI: 10.1243/0309324981512841.
- [2] G. Pitarresi and E. A. Patterson. **“A review of the general theory of thermoelastic stress analysis”**. In: *The Journal of Strain Analysis for Engineering Design*, 38(5), 405-417 (2003). DOI: 10.1243/03093240360713469.
- [3] K. Nishioka et al. **“Development of defect localization method for perforated carbon-fiber-reinforced plastic specimens using finite element method and graph neural network”**. In: *Frontiers in Materials*, 12, 1652484 (2025). DOI: 10.3389/fmats.2025.1652484.
- [4] Q. Hernandez et al. **“Thermodynamics-informed graph neural networks”**. In: *IEEE Transactions on Artificial Intelligence* (2022). DOI: 10.1109/TAI.2022.3179681. arXiv: 2203.01874.
- [5] Q. Hernandez et al. **“Structure-preserving neural networks”**. In: *Journal of Computational Physics*, vol. 426, 109950 (2021). DOI: 10.1016/j.jcp.2020.109950. arXiv: 2004.04653.
- [6] M. Tancik et al. **“Fourier Features Let Networks Learn High Frequency Functions in Low Dimensional Domains”**. In: *NeurIPS 2020*. 2020. arXiv: 2006.10739.
- [7] Y. Rong et al. **“DropEdge: Towards Deep Graph Convolutional Networks on Node Classification”**. In: *ICLR 2020*. 2020. arXiv: 1907.10903.

- [8] T.-Y. Lin et al. **“Focal Loss for Dense Object Detection”**. In: *IEEE International Conference on Computer Vision (ICCV)*, pp. 2980-2988. 2017. DOI: 10.1109/ICCV.2017.324. arXiv: 1708.02002.
- [9] K. Cao et al. **“Learning Imbalanced Datasets with Label-Distribution-Aware Margin Loss”**. In: *Advances in Neural Information Processing Systems (NeurIPS 2019)*, vol. 32. 2019. arXiv: 1906.07413.
- [10] A. K. Menon et al. **“Long-tail learning via logit adjustment”**. In: *International Conference on Learning Representations (ICLR 2021)*; *arXiv preprint 2020*. 2020. arXiv: 2007.07314.
- [11] T. Zhao et al. **“GraphSMOTE: Imbalanced Node Classification on Graphs with Graph Neural Networks”**. In: *14th ACM International Conference on Web Search and Data Mining (WSDM)*. 2021. DOI: 10.1145/3437963.3441720. arXiv: 2103.08826.
- [12] P. Velickovic et al. **“Graph Attention Networks”**. In: *International Conference on Learning Representations (ICLR 2018)*. 2018. arXiv: 1710.10903.
- [13] S. Brody et al. **“How Attentive are Graph Attention Networks? (GATv2)”**. In: *International Conference on Learning Representations (ICLR 2022)*. 2022. arXiv: 2105.14491.
- [14] T. Pfaff et al. **“Learning Mesh-Based Simulation with Graph Networks”**. In: *International Conference on Learning Representations (ICLR 2021)*. 2021. arXiv: 2010.03409.

- [15] A. Sanchez-Gonzalez et al. **“Learning to Simulate Complex Physics with Graph Networks”**. In: *ICML 2020 (PMLR Proceedings of the 37th ICML)*. 2020. arXiv: 2002.09405.
- [16] V. G. Satorras et al. **“E(n) Equivariant Graph Neural Networks (EGNN)”**. In: *ICML 2021 (PMLR 139, pp. 9323-9332)*. 2021. arXiv: 2102.09844.